

CHARLOTTE DOUGLAS, NC, USA

WMO: 723140

Lat: 35.224N

Lon: 80.955W

Elev: 222

StdP: 98.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13881

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.1	-4.0	-16.9	0.9	-1.0	-14.5	1.1	-0.3	10.1	11.7	8.5	9.0	2.2	310	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	34.6	23.7	33.3	23.4	32.1	23.1	25.2	31.3	24.7	30.4	24.1	29.6	2.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.6	18.9	27.3	23.0	18.3	26.8	22.6	17.8	26.4	77.8	31.4	75.6	30.6	73.6	29.7	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.2	6.1	-10.1	36.1	2.3	1.7	-11.8	37.4	-13.2	38.4	-14.5	39.3	-16.2	40.6		
(5)			-11.3	26.1	2.1	0.7	-12.8	26.6	-14.0	27.0	-15.2	27.4	-16.7	27.9	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.4	5.5	7.6	11.5	16.3	20.7	24.8	26.4	25.9	22.6	16.6	10.8	7.3	(6)	(7)
	DBStd	8.42	5.49	5.12	5.07	4.22	3.56	2.69	2.07	2.18	3.16	4.22	4.52	4.91	(7)	(8)
	HDD10.0	468	159	98	44	4	0	0	0	0	0	3	44	115	(8)	(9)
	HDD18.3	1683	397	301	218	87	20	1	0	0	6	82	229	343	(9)	(10)
	CDD10.0	2795	21	32	90	194	331	443	507	492	377	209	68	31	(10)	(11)
	CDD18.3	968	0	1	6	26	92	194	249	233	133	29	2	1	(11)	(12)
	CDH23.3	8738	1	6	54	251	787	1802	2412	2146	1043	222	12	2	(12)	(13)
	CDH26.7	3414	0	0	6	48	236	746	1074	902	360	43	1	0	(13)	(14)
(14) Wind	WSAvg	2.7	2.8	3.1	3.3	3.3	2.9	2.6	2.4	2.3	2.6	2.4	2.5	2.5	(14)	(15)
(15) Precipitation	PrecAvg	1096	88	78	102	87	78	104	106	104	97	83	79	91	(15)	(16)
	PrecMax	1487	167	145	210	212	243	197	225	230	203	348	240	221	(16)	(17)
	PrecMin	734	24	28	32	22	4	21	44	13	8	0	9	33	(17)	(18)
	PrecStd	178	37	30	46	47	50	52	44	53	45	70	52	43	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.6	24.2	27.4	30.1	32.8	35.8	36.1	35.8	33.8	30.3	25.0	22.8	(19)	(20)
		MCWB	16.2	16.6	17.0	19.0	22.1	23.4	23.9	23.5	21.9	21.6	17.2	16.8	(20)	(21)
	2%	DB	19.0	21.0	24.9	28.2	30.9	33.8	34.5	34.1	32.0	27.6	22.7	19.9	(21)	(22)
		MCWB	14.7	14.3	15.9	18.3	21.1	23.4	24.1	23.6	22.0	19.8	16.0	16.1	(22)	(23)
	5%	DB	17.0	18.7	22.6	26.3	29.3	32.3	33.2	32.7	30.4	25.8	20.8	17.7	(23)	(24)
		MCWB	13.3	13.4	14.5	17.2	20.8	23.0	23.7	23.6	21.7	18.7	15.7	14.5	(24)	(25)
	10%	DB	14.5	16.5	20.1	24.4	27.7	30.8	31.9	31.2	28.8	24.0	18.9	15.7	(25)	(26)
		MCWB	10.7	11.4	13.1	16.5	19.9	22.4	23.4	23.1	21.3	18.3	14.6	12.3	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.0	18.3	18.9	21.3	23.9	25.2	26.0	25.9	24.7	24.0	20.2	19.0	(27)	(28)
		MCDB	19.7	21.7	24.4	26.8	29.4	32.1	32.9	31.8	29.4	26.7	21.4	20.8	(28)	(29)
	2%	WB	16.1	16.5	17.4	20.0	22.7	24.5	25.2	25.2	23.9	22.0	18.4	17.1	(29)	(30)
		MCDB	18.2	19.6	22.2	25.2	28.2	30.9	31.6	31.0	28.4	25.0	20.9	19.0	(30)	(31)
	5%	WB	14.0	14.5	16.2	18.9	21.8	23.9	24.7	24.6	23.1	20.6	17.1	15.2	(31)	(32)
		MCDB	16.5	17.6	20.4	23.8	27.2	30.0	30.7	30.0	27.1	23.9	19.6	17.2	(32)	(33)
	10%	WB	11.3	12.2	14.9	17.8	20.9	23.2	24.2	24.0	22.4	19.3	15.3	12.9	(33)	(34)
		MCDB	13.8	15.2	18.7	22.5	25.9	29.0	29.8	29.0	26.4	22.7	18.1	14.7	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	11.0	11.8	12.2	11.3	10.6	10.3	10.2	10.3	11.6	11.9	10.3	(35)	(36)
		MCDBR	12.7	13.5	14.2	14.0	12.7	12.4	12.0	11.8	12.4	13.0	13.8	11.8	(36)	(37)
	5% WB	MCWBR	8.7	8.6	7.4	6.4	5.1	4.2	3.8	3.6	4.4	6.1	8.1	8.3	(37)	(38)
		MCDWR	11.0	11.5	11.9	11.4	11.1	10.8	10.6	10.2	9.8	9.7	10.2	9.9	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub	taub	0.313	0.322	0.360	0.403	0.458	0.485	0.536	0.517	0.428	0.380	0.330	0.312	(39)	(40)
		taud	2.498	2.469	2.397	2.270	2.154	2.117	1.999	2.038	2.273	2.385	2.486	2.533	(40)	(41)
	Ebn at Noon	Ebn at Noon	893	923	911	883	835	809	764	771	829	843	863	872	(41)	(42)
		Edh at Noon	83	95	112	134	152	158	176	167	125	103	84	76	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg	RadAvg	2.64	3.29	4.35	5.44	6.04	6.42	6.00	5.49	4.66	3.84	2.91	2.27	(43)	(44)
	RadStd	RadStd	0.15	0.38	0.31	0.42	0.49	0.46	0.40	0.26	0.41	0.47	0.24	0.21	(44)	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (69 neighbors)	+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140	+117
		+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+162	+110

Nomenclature: See separate page